

Power BI Sales Analysis Report
Mission X Data Analytics Project

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Course:
Data Analytics Launchpad

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Mission Ready

1. Executive Summary

This report analyses e-commerce sales data using Power BI to identify trends in revenue, products, customer behaviour. Data was cleansed, modelled, and enhanced with DAX measures such as Total Sales and YTD comparisons, while a secondary dataset provided insights into revenue by customer demographics like age group and gender, provides a border context and support to the e-commerce sales data.

Key findings include:

Total Sales: \$1,911,799.86, with the highest monthly revenue of \$113,914.85 in October 2016.

Top Products: The “Uncategorised” category (\$190,561.81) and Christmas RETROSPOT STAR WOOD (\$77,183.60) drove significant revenue.

Customer Insights: The UK accounted for the largest customer base at 90.73%(73,964), the 51-60 age group generated the most revenue (\$45,316) and the most previous purchases (19935), and Male customers show dominance in gender in both total revenue (67.74%) and total previous purchases (68.95%).

Forecasting: A 90-day forecast aligns with historical trends, aiding short-term planning.

Recommendations include focusing on high-performing products, leveraging seasonal trends, targeting key regions, and using demographic insights for marketing strategies.

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2. Introduction

This report presents the analysis completed for the Mission X Data Analytics project. The purpose of the report is to demonstrate how data cleansing, SQL queries, Power BI visualisation, and DAX measures were used to generate meaningful insights from a sales dataset.

The dataset used in this project includes information about customer demographics, product details, invoice transactions, and sales performance over time. After cleaning and preparing the data with Power Query in Power BI, several visualisations were created to explore trends such as monthly sales performance, top-selling products, customer distribution by country, product category performance, and a 90-day sales forecast. In addition, a Date table was created and linked to the Sales table to support time intelligence analysis.

This report summarises the key steps taken in the analysis process, explains the techniques used, and highlights the insights gained to support data-driven decision-making.

3. Methodology

This section outlines the process used to prepare, clean, model, and analyse the sales and shopping behaviours datasets for this project. The methodology followed the structured data analytics lifecycle: data connection, data cleansing, data modelling, calculation creation, and visualisation.

3.1 Connecting to the Database

Power BI was connected directly to the Azure SQL Database using the provided server and login credentials as well as a self-sourced data set. The following tables were imported:

- Sales
- Invoice
- Product
- Shopping behaviours dataset (Kaggle)

These tables formed the foundation of the analysis and were loaded into Power BI for transformation and modelling.

3.2 Data Cleansing Using Power Query

All data cleaning was performed in Power Query to ensure the dataset was accurate, consistent, and ready for analysis. Key transformations included:

- Removing duplicate invoice records to avoid double-counting
- Handling missing or incorrect values in fields such as product descriptions and categories
- Standardising text fields, such as ensuring country names were consistent (e.g., “AUS” changed to “Australia”)
- Converting InvoiceDate into a proper date format to support time-based visuals
- Checking and correcting data types, ensuring numbers, dates, and text fields were formatted correctly
- Creating a Sales measure column (Quantity × UnitPrice)

No SQL cleansing was required, as all cleaning took place directly within Power BI.

After cleansing, the data was loaded into the model for relationship creation.

3.3 Data Modelling in Power BI

The data model was built in Model View, where relationships were created between tables using their common keys:

- Sales → Invoice joined on InvoiceNo
- Sales → Product joined on StockCode
- Sales → Date joined on InvoiceDate/Date

These relationships enabled the tables to interact correctly, allowing calculations and visuals to reflect accurate combined insights across sales, products, and countries.

A separate Date table was also created to support time-intelligence features such as YTD and last-year comparisons.

3.4 Calculated Columns and Measures

Power BI DAX was used to create both calculated columns and measures:

- Calculated Column:
Sales = Quantity × UnitPrice
- Measures:
 - Total Sales = SUM(Sales, Sales[Quantity] * Sales[UnitPrice])
 - TOTALYTD TOTALYTD = TOTALYTD([Total Sales Measure]
'Date'[Date])
 - TOTALYTD Last Year using SAMEPERIODLASTYEAR =
TOTALYTD_LastYear=CALCULATE([TOTALYTD], SAMEPERIODLAST
YEAR('Date'[Date]))

These measures were used across the visuals to produce accurate numerical insights.

3.5 Data Visualisation

A range of Power BI visuals were created to answer key business questions:

- Line chart: Total Sales over time
- Bar charts:
 - Top product categories
 - Top-selling products
- Bar chart: Customers by country
- Table: Month, Year, Total Sales, YTD Sales, Last Year YTD
- Forecast visual: 90-day daily sales forecast
- Clustered Column Chart: Revenue by Age Group and Gender

Bookmarks were added to create a guided, storytelling flow and improve navigation for the report audience.

4. Analysis & Insights

4.1 Overview

The analysis focused on identifying key sales trends, product performance, and customer behaviour using the primary Sales dataset. Power BI was the primary tool for data modelling, transformation, and visualisation, with the secondary dataset providing additional context for demographic and market analysis.

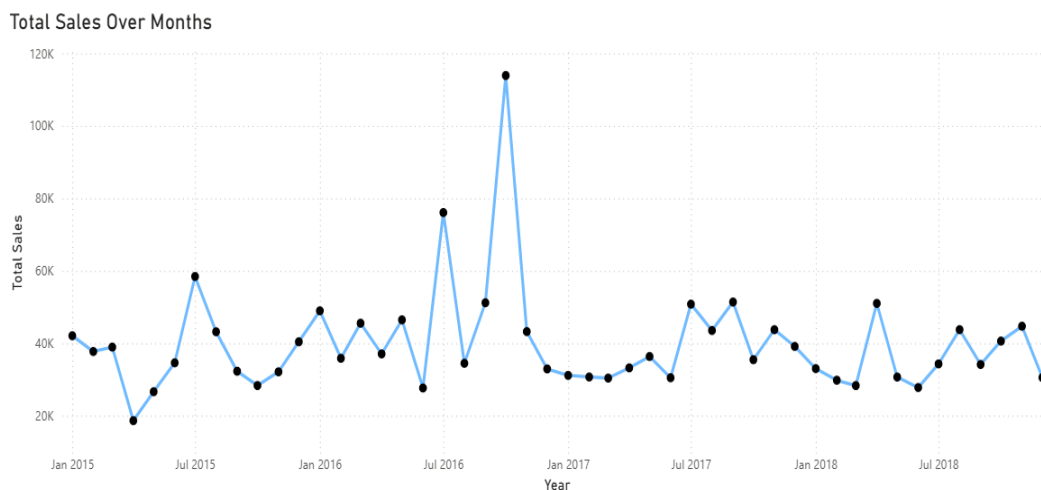
4.2 Data Visualisations

Several visualisations were created to highlight insights:

- Sales Over Time: Line chart displaying monthly sales trends, revealing seasonality and growth patterns.
- Top Categories: Bar chart showing sales by product category to identify the most popular product groups.
- Top Products: Bar chart highlighting individual high-performing products.
- Customer Demographics: Bar charts showing countries with the largest customer base and total revenue by Age Group, providing insights into international reach.
- YTD Sales Table: Table including Month, Year, Total Sales, TOTALYTD, and TOTALYTD_LastYear for easy comparison of sales trends over time.
- 90-Day Forecast: Daily forecast visual to predict future sales based on historical trends and seasonality.

4.3 Key Insights

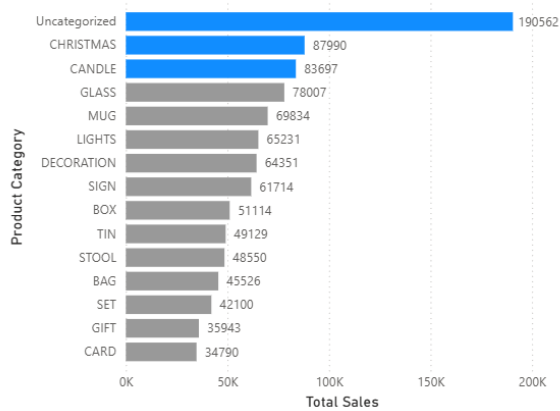
- Overall Sales Performance



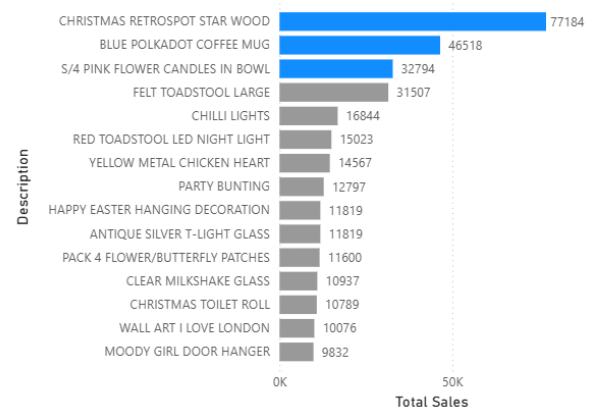
Sales exhibited clear monthly peaks, with October 2016 reaching the highest sales of \$113,914.85 and July 2016 reaching the second highest sales of \$76,089.78. A pattern shown in July used to have a higher total sale in every year, suggesting seasonal demand and opportunities for targeted promotions during high-sales months.

- **Top-Selling Categories and Products**

Total Sales by Product Category



Total Sales by Products

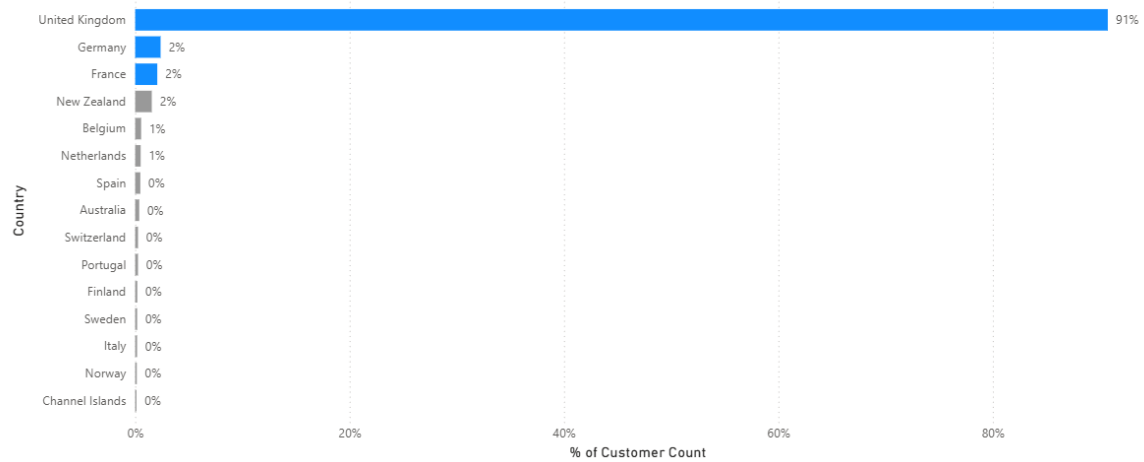


The Uncategorized product category generated the highest total sales at \$190,561.81, showing a strong lead over all other categories and indicating that products outside standard groupings contributed significantly to overall revenue. The next highest category was Christmas, with \$87,990, highlighting a substantial gap between the top category and the rest.

The top selling product was the Christmas Retrospot Star Wood with \$77,183.60, followed by the Blue Polkadot Coffee Mug at \$46,518. This shows that while a few standout products drive large portions of revenue, there is also a noticeable drop-off between the highest and second-highest selling items.

- **Regional Performance**

% of Customer Count by Country



The United Kingdom accounted for around 91% of all customers, totalling 73,964 individuals. This shows a strong dominance over the other countries, indicating the UK as the primary market for targeted growth and marketing opportunities.

- **Time Intelligence Analysis:**

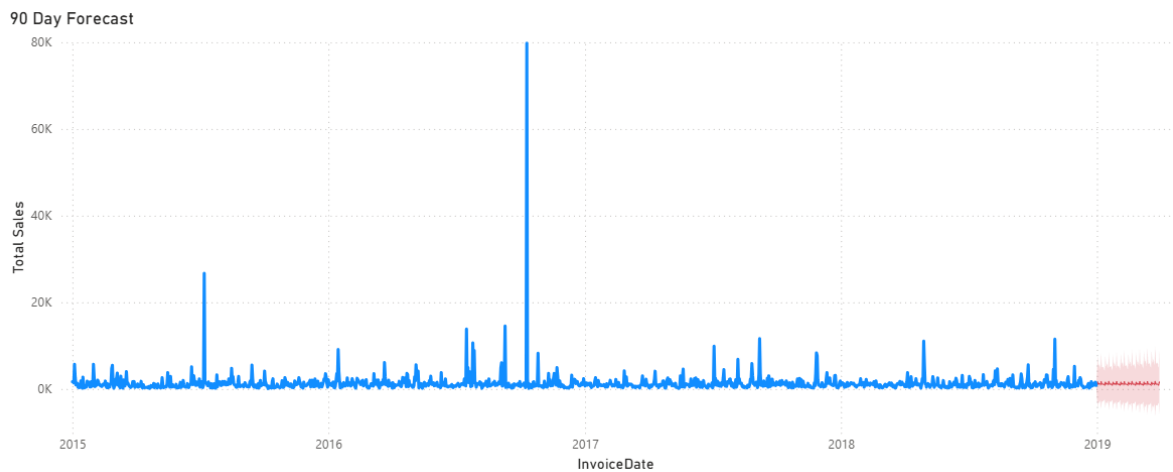
Year	Month	Total Sales	TOTALYTD	TOTALYTD_LastYear
2015	January	42,071.47	42,071.47	
2015	February	37,727.10	79,798.57	
2015	March	38,946.36	118,744.93	
2015	April	18,638.58	137,383.51	
2015	May	26,631.63	164,015.14	
2015	June	34,611.77	198,626.91	
2015	July	58,438.22	257,065.13	
2015	August	43,178.18	300,243.31	
2015	September	32,279.50	332,522.81	
2015	October	28,332.50	360,855.31	
2015	November	32,118.92	392,974.23	
2015	December	40,422.50	433,396.73	
2016	January	48,962.35	48,962.35	42,071.47
2016	February	35,856.58	84,818.93	79,798.57
2016	March	45,540.09	130,359.02	118,744.93
2016	April	37,064.80	167,423.82	137,383.51
2016	May	46,470.12	213,893.94	164,015.14
2016	June	27,654.55	241,548.49	198,626.91
2016	July	76,089.78	317,638.27	257,065.13
2016	August	34,479.95	352,118.22	300,243.31
2016	September	51,189.26	403,307.48	332,522.81
2016	October	113,914.85	517,222.33	360,855.31
2016	November	43,230.93	560,453.26	392,974.23
2016	December	32,920.51	593,373.77	433,396.73
2017	January	31,129.91	31,129.91	48,962.35
2017	February	30,689.05	61,818.96	84,818.93
2017	March	30,388.88	92,207.84	130,359.02
2017	April	33,213.68	125,421.52	167,423.82
2017	May	36,353.92	161,775.44	213,893.94
Total		1,911,799.86	428,681.75	456,347.61

The Year-To-Date (YTD) and prior-year comparisons reveal important trends:

- Total Sales (Sum of all sales): \$1,911,799.86
- YTD Sales: \$428,681.75
- YTD Last Year: \$456,347.61

This indicates a slight decrease in YTD sales compared to the previous year, highlighting areas for potential growth. Using the Date table and DAX measures like TOTALYTD and SAMEPERIODLASTYEAR, trends over time were clearly tracked and analysed.

- **Growth and Forecasting:**

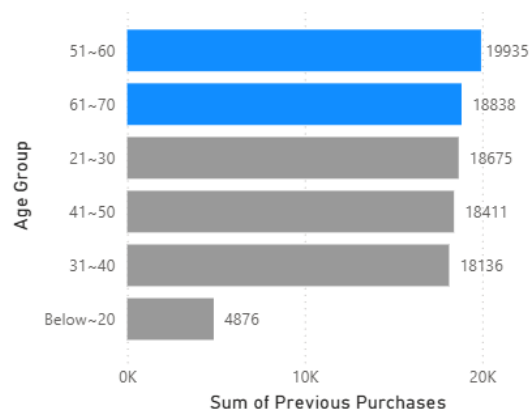


The 90-day sales forecast projected future trends based on historical data and seasonality, enabling proactive planning for inventory and sales strategies. The forecast shows consistent patterns aligned with previous monthly trends, providing actionable insights for short-term decision-making.

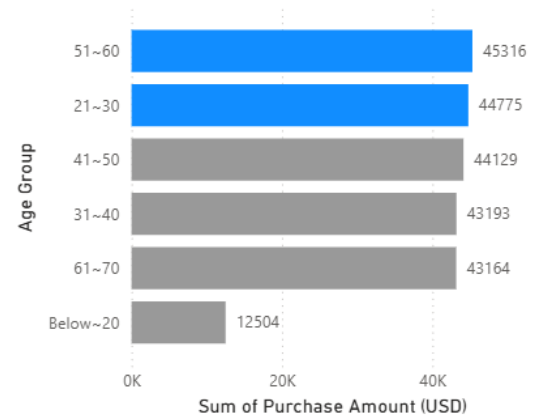
5. Secondary Dataset Context

For the secondary dataset, age and gender are the customer demographics used to analyse the total revenue.

Total Previous Purchases by Age Group

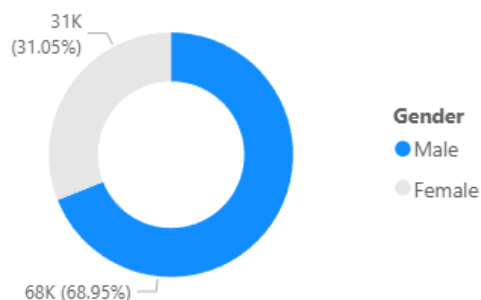


Total Purchase Amount (USD) by Age Group

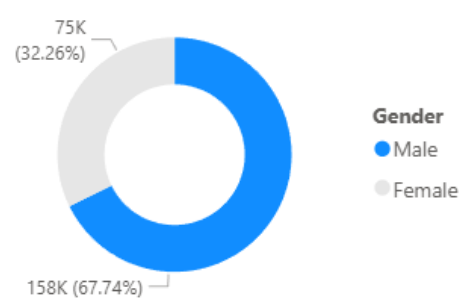


Firstly, for the age group, 51-60 is the age group generated the highest revenue at \$45316 and has the highest number of previous purchases than other age groups at 19935, while the age group below or at 20 had the lowest revenue at \$12504 and the lowest previous purchases at 4876. This indicates that older customers, especially the age group between 51 and 60, contributed more to overall sales, which could inform marketing or product strategies targeting these age groups and provides a border context for the insights derived from primary dataset to understand customer demographics in age group.

Total Previous Purchases by Gender



Total Purchase Amount (USD) by Gender



For the gender, Male is the gender which shows dominance in both total revenue (67.74%) and total previous purchases (68.95%). This shows that male customers spend more overall than female customers, contributing a larger share of total sales. This insight could help guide marketing or product strategies aimed at engaging male

buyers, provides a border context for the insights derived from primary dataset to understand customer demographics in gender.

6. Conclusions

Based on the analysis conducted:

- **Total Sales Performance:** The dataset recorded total sales of \$1,911,799.86, with the highest monthly sales in October 2016 reaching \$113,914.85, indicating clear seasonal peaks.
- **Top Categories and Products:** The “Uncategorised” category generated \$190,561.81, while the top-selling product, CHRISTMAS RETROSPOT STAR WOOD, achieved \$77,183.60, highlighting which products and categories drive revenue.
- **Customer Demographics:** Around 91% of customers were in the United Kingdom (73,964 customers), identifying the key market for targeted marketing and growth strategies.
- **Time Intelligence Insights:** YTD sales were \$428,681.75, slightly lower than the previous year’s YTD of \$456,347.61, highlighting a need to investigate factors affecting current-year performance.
- **Forecasting:** The 90-day sales forecast demonstrated consistent trends aligned with historical data, providing actionable insights for inventory planning, sales strategy, and short-term revenue management.
- **Secondary Data Set-** Male customers and the 51–60 age group contributed the most to total revenue, reinforcing how demographics influence purchasing behaviour and helping identify key target audiences.

These data-driven conclusions provide a clear, actionable summary of sales trends, customer behaviour, and product performance, supporting informed decision-making for business strategy.

7. Recommendations

Based on the analysis and actual data, the following actions are recommended:

1. Focus on High-Performing Products and Categories:
 - Prioritise marketing and inventory for the “Uncategorised” category (\$190,561.81 in sales) and the top-selling product Christmas RETROSPOT STAR WOOD (\$77,183.60), as these contributed disproportionately to revenue.
2. Leverage Seasonal Trends:
 - Plan targeted promotions during peak sales months, especially October 2016, which recorded \$113,914.85 in sales, to maximise revenue during high-demand periods.
3. Target Key Regions:
 - Focus marketing and sales strategies on the United Kingdom, which accounted for 73,964 customers, representing the largest customer base.
4. Monitor Sales Using Dashboards:
 - Continue using Power BI dashboards to track YTD sales (\$428,681.75) versus previous year (\$456,347.61), identify trends early, and support proactive inventory and sales decisions.
5. Integrate Secondary Dataset Insights:
 - Regularly incorporate demographic and market trend data to benchmark performance against the broader e-commerce industry and identify growth opportunities.
 - Target age group between 51 to 60 and Male gender derived from the secondary dataset insights.
 - Investigate Lower-Performing Segments: Analyse why other age groups or female customers spend less, this may uncover product preferences, accessibility issues, or engagement gaps.